

MakonBook Structured PDF v2

Deterministic transport format for EBRW / Math imports with automatic visual extraction

What changed in v2

- **No AI layout guessing inside MakonBook.** Question fields remain selectable marker text.
- **Tables, graphs and diagrams stay visual.** They are placed between exact VISUAL markers and cropped locally to PNG.
- **Question counts remain dynamic.** Placement and custom tests can contain any number of questions.
- **v1 remains accepted.** Existing staging imports do not need to be rebuilt just because v2 exists.

Required header

```
[[MAKONBOOK_STRUCTURED_PDF:2]]  
[[SECTION:EBRW]]  
[[MODULE:1]]
```

For Math use `[[SECTION:MATH]]`. Module 2 begins with `[[MODULE:2]]`.

Question without a visual

```
[[QUESTION:1]]
[[TYPE:MCQ]]
[[PASSAGE]]
Passage text...
[/PASSAGE]]
[[PROMPT]]
Question prompt...
[/PROMPT]]
[[A]]Choice A[/A]]
[[B]]Choice B[/B]]
[[C]]Choice C[/C]]
[[D]]Choice D[/D]]
[[ANSWER]]B[/ANSWER]]
[[EXPLANATION]][/EXPLANATION]]
[[END_QUESTION]]
```

Question with a table / chart / diagram

```
[[PASSAGE]]
[[VISUAL:QUESTION]]
<place ONLY the required cropped visual here>
[/VISUAL:QUESTION]]
```

```
Source text that belongs to the passage...
[/PASSAGE]]
```

Important: the cropped visual must not include the question header, answer choices, selected-answer highlight, watermark, or unrelated surrounding content. The two VISUAL marker lines must be selectable text and must stay on the same PDF page.

Example: preserve a table as an image

The source question contains a table. Do not flatten the table into a sentence such as "Belgium 4 13 11 Italy 14 57 42...". Keep the table visual and keep the explanatory paragraph as text.

[[VISUAL:QUESTION]]

Country	Trees	Fungi	Insects
Belgium	4	13	11
Italy	14	57	42
Denmark	12	22	33

[/VISUAL:QUESTION]

Choice visuals (Math)

```
[[A]]
[[VISUAL:A]]
<only choice A visual>
[/VISUAL:A]]
[/A]]
```

Use VISUAL:B, VISUAL:C and VISUAL:D the same way. MakonBook maps those crops directly to image_a, image_b, image_c and image_d.

Validation

- A declared visual whose crop cannot be found becomes a blocking validation error.
- A main visual is attached automatically to the staging question image field.
- Multiple numeric main visuals (VISUAL:1, VISUAL:2, ...) are combined vertically into one question image.
- Published test editing and deletion are restricted to Manager/Admin authorization.